
Quantum Physics

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Assignment: Number 2

Problem 1 Derive the wave function of a 1D infinite square well

$$V(x) = \begin{cases} 0, & |x| \leq a \\ \infty, & |x| > a \end{cases}$$

(a) 1D time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Since $V = \infty$ when $|x| > a$, the term $V(x)\psi(x)$ in the Schrödinger equation would diverge unless $\psi(x) = 0$. To keep the Schrödinger equation finite and well-defined, the wave function must vanish in this region:

$$\psi(x) = 0 \quad (|x| > a)$$

which means the particle cannot exist outside the region $[-a, a]$.

The wave function has physical meaning only inside the region $[-a, a]$, so we solve the Schrödinger equation in this interval.

(b) Solve the Schrödinger equation

Inside the well $V(x) = 0$, therefore

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

which can be written as

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

The general solution of this differential equation is

$$\psi(x) = C_1 e^{\frac{i\sqrt{2mE}}{\hbar}x} + C_2 e^{-\frac{i\sqrt{2mE}}{\hbar}x}$$

Since the probability density must be continuous in space, the wave function must also be continuous. Outside the well the wave function is zero, therefore at the boundaries we must have

$$\psi(-a) = 0, \quad \psi(a) = 0$$

Substituting the boundary conditions into the general solution gives

$$C_1 e^{\frac{i\sqrt{2mE}}{\hbar}a} + C_2 e^{-\frac{i\sqrt{2mE}}{\hbar}a} = 0$$

$$C_1 e^{-\frac{i\sqrt{2mE}}{\hbar}a} + C_2 e^{\frac{i\sqrt{2mE}}{\hbar}a} = 0$$

Adding and subtracting these two equations, we obtain

$$2(C_1 + C_2) \cos\left(\frac{\sqrt{2mE}}{\hbar}a\right) = 0$$

$$2(C_1 - C_2) \sin\left(\frac{\sqrt{2mE}}{\hbar}a\right) = 0$$

Since cos and sin cannot be zero at the same time, there are two cases.

- If

$$\cos\left(\frac{\sqrt{2mE}}{\hbar}a\right) = 0$$

then $C_1 = C_2$, and

$$\frac{\sqrt{2mE}}{\hbar}a = \frac{2n+1}{2}\pi$$

The wave function becomes

$$\psi(x) = 2C_1 \cos\left(\frac{2n+1}{2} \frac{\pi x}{a}\right)$$

- If

$$\sin\left(\frac{\sqrt{2mE}}{\hbar}a\right) = 0$$

then $C_1 = -C_2$, and

$$\frac{\sqrt{2mE}}{\hbar}a = \frac{2n\pi}{2}$$

The wave function becomes

$$\psi(x) = 2C_2 \sin\left(\frac{2n\pi x}{2a}\right)$$

Notice that these two forms can be written in a unified expression

$$\psi(x) = A \sin \left(\frac{n\pi x}{2a} + \frac{n\pi}{2} \right), \quad n \in \mathbb{N}$$

and the corresponding energy eigenvalues are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8a^2 m}.$$

(c) Normalization

The normalization condition is

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

Since the wave function is nonzero only in $[-a, a]$,

$$\int_{-a}^a |\psi(x)|^2 dx = 1$$

$$A^2 \int_{-a}^a \sin^2 \left(\frac{n\pi x}{2a} + \frac{n\pi}{2} \right) dx = 1$$

$$A^2 a = 1$$

Therefore

$$A = \sqrt{\frac{1}{a}}$$

Thus the normalized spatial wave function is

$$\psi_n(x) = \sqrt{\frac{1}{a}} \sin \left(\frac{n\pi x}{2a} + \frac{n\pi}{2} \right)$$

The time dependent factor is

$$e^{-iE_n t/\hbar}$$

Therefore the full wave function is

$$\Psi_n(x, t) = \sqrt{\frac{1}{a}} \sin \left(\frac{n\pi x}{2a} + \frac{n\pi}{2} \right) e^{-iE_n t/\hbar}$$

Problem 2 Quantum-to-Classical Correspondence

- (a) Similar to Young's double-slit interference experiment, quantum effects become important when the wavelength is comparable to the system size. When the wavelength is much smaller than the experimental scale, the behavior approaches the classical limit.

For the infinite square well, the wave number is

$$k = \frac{n\pi}{2a}$$

so the wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{4a}{n}.$$

When $n \rightarrow \infty$, the wavelength becomes very small compared with the well size. In this limit the wave function oscillates very rapidly.

The probability density is

$$|\Psi(x, t)|^2 = \frac{1}{a} \sin^2 \left(\frac{n\pi(x + a)}{2a} \right).$$

For a small measurement interval δ ,

$$\langle |\psi(x)|^2 \rangle = \frac{\int_{x-\delta/2}^{x+\delta/2} \frac{1}{a} \sin^2(kx) dx}{\int_{x-\delta/2}^{x+\delta/2} dx}.$$

We use the identity

$$\sin^2(kx) = \frac{1}{2} - \frac{\cos(2kx)}{2}.$$

Because the oscillation of $\cos(2kx)$ is very fast as $k \rightarrow \infty$, the integral of the second term over a small interval is approximately zero. Therefore the integral is determined by the first term. Over the interval $[x - \frac{\delta}{2}, x + \frac{\delta}{2}]$,

$$\int_{x-\frac{\delta}{2}}^{x+\frac{\delta}{2}} \frac{1}{a} \sin^2(kx) dx \approx \int_{x-\frac{\delta}{2}}^{x+\frac{\delta}{2}} \frac{1}{2a} dx = \frac{\delta}{2a}.$$

Substituting into the average probability density,

$$\langle |\psi(x)|^2 \rangle = \frac{\frac{\delta}{2a}}{\delta} = \frac{1}{2a}.$$

Thus the probability density becomes approximately constant in the interval $[-a, a]$. This agrees with the classical expectation that the particle can appear with equal probability anywhere inside the well.

- (b) Consider a particle moving in this potential field. We can represent the infinite potential well using the step function

$$U(q) = \lim_{h \rightarrow \infty} (h - h u(q + a) + h u(q - a)).$$

The Lagrangian is

$$\mathcal{L} = \frac{1}{2}m\dot{q}^2 - \lim_{h \rightarrow \infty} (h - h u(q + a) + h u(q - a)).$$

The generalized momentum is

$$p = \frac{\partial \mathcal{L}}{\partial \dot{q}} = m\dot{q}.$$

Then the Hamiltonian is

$$\mathcal{H} = \sum p\dot{q} - \mathcal{L} = \frac{1}{2}m\dot{q}^2 + \lim_{h \rightarrow \infty} (h - h u(q + a) + h u(q - a)).$$

When q is outside the interval $[-a, a]$, the generalized energy approaches

$$\mathcal{H} \rightarrow \infty,$$

so the motion has no physical meaning outside the well.

The equation of motion is

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = 0$$

$$m\ddot{q} - h\delta(q + a) + h\delta(q - a) = 0.$$

Let

$$\ddot{q} = \dot{q} \frac{d\dot{q}}{dq}.$$

Then

$$m\dot{q} d\dot{q} = (h\delta(q + a) - h\delta(q - a))dq.$$

Integrating both sides gives

$$\frac{1}{2}m\dot{q}^2 = C + h u(q + a) - h u(q - a).$$

Inside the interval $[-a, a]$, the kinetic energy is constant,

$$\frac{1}{2}m\dot{q}^2 = K.$$

Therefore

$$|\dot{q}| = \text{constant}.$$

Thus the particle moves with constant speed inside the well and only changes the direction of its velocity at the boundaries $q = \pm a$. This corresponds to a uniform probability density, since the particle spends equal time in equal spatial intervals inside the well.

(c) Sommerfeld Quantization Condition

We now apply the Sommerfeld quantization condition to a particle moving inside the box of length $2a$ with constant speed.

$$p = m\dot{q}.$$

The quantization condition is

$$\oint p \, dq = nh.$$

For one complete classical period, the particle travels from $-a$ to a and back to $-a$, so the total distance is $4a$. Therefore

$$\oint p \, dq = m\dot{q} \cdot 4a = 4am\dot{q} = nh.$$

Thus

$$\dot{q} = \frac{nh}{4am}.$$

The classical energy is

$$E = \frac{1}{2}m\dot{q}^2 = \frac{1}{2}m \left(\frac{nh}{4am} \right)^2.$$

Therefore

$$E_n = \frac{n^2 h^2}{32 a^2 m} = \frac{n^2 \pi^2 \hbar^2}{8 a^2 m}.$$

This result is exactly the same as the energy eigenvalues obtained from solving the Schrödinger equation.